

AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus

E7.1 - Arc Length

(A-Level)

Name: _____

Class: _____

Date: _____

Time: 160 min.

Marks: 134 marks

Comments:

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Q1.

A curve is defined parametrically by

$$x = t^3 + 5, \quad y = 6t^2 - 1$$

The arc length between the points where $t = 0$ and $t = 3$ on the curve is s .

(a) Show that $s = \int_0^3 3t\sqrt{t^2 + A} \, dt$, stating the value of the constant A .

(4)

(b) Hence show that $s = 61$.

(4)

(Total 8 marks)

Q2.

A curve has cartesian equation

$$y = \frac{1}{2} \ln(\tanh x)$$

(a) Show that

$$\frac{dy}{dx} = \frac{1}{\sinh 2x}$$

(4)

(b) The points A and B on the curve have x -coordinates $\ln 2$ and $\ln 4$ respectively. Find the arc length AB , giving your answer in the form $p \ln q$, where p and q are rational numbers.

(8)

(Total 12 marks)

Q3.

(a) Given that

$$x = \ln(\sec t + \tan t) - \sin t$$

show that

$$\frac{dx}{dt} = \sin t \tan t$$

(4)

(b) A curve is given parametrically by the equations

$$x = \ln(\sec t + \tan t) - \sin t, \quad y = \cos t$$



The length of the arc of the curve between the points where $t = 0$ and $t = \frac{\pi}{3}$ is denoted by s .

Show that $s = \ln p$, where p is an integer.

(6)

(Total 10 marks)

Q4.

- (a) Using the identities

$$\cosh^2 t - \sinh^2 t = 1, \tanh t = \frac{\sinh t}{\cosh t} \text{ and } \operatorname{sech} t = \frac{1}{\cosh t}$$

show that:

(i) $\tanh^2 t + \operatorname{sech}^2 t = 1$;

(2)

(ii) $\frac{d}{dt}(\tanh t) = \operatorname{sech}^2 t$;

(3)

(iii) $\frac{d}{dt}(\operatorname{sech} t) = -\operatorname{sech} t \tanh t$.

(3)

- (b) A curve C is given parametrically by

$$x = \operatorname{sech} t, y = 4 - \tanh t$$

- (i) Show that the arc length, s , of C between the points where $t = 0$ and $t = \frac{1}{2} \ln 3$ is given by

$$s = \int_0^{\frac{1}{2} \ln 3} \operatorname{sech} t \, dt$$

(4)

- (ii) Using the substitution $u = e^t$, find the exact value of s .

(6)

(Total 18 marks)

Q5.

- (a) Show that

$$\frac{d}{dx} \left(\cosh^{-1} \frac{1}{x} \right) = \frac{-1}{x\sqrt{1-x^2}}$$

(3)

(b) A curve has equation

$$y = \sqrt{1-x^2} - \cosh^{-1} \frac{1}{x} \quad (0 < x < 1)$$

Show that:

(i) $\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{x}$;

(4)

(ii) the length of the arc of the curve from the point where $x = \frac{1}{4}$ to the point where $x = \frac{3}{4}$ is $\ln 3$.

(5)

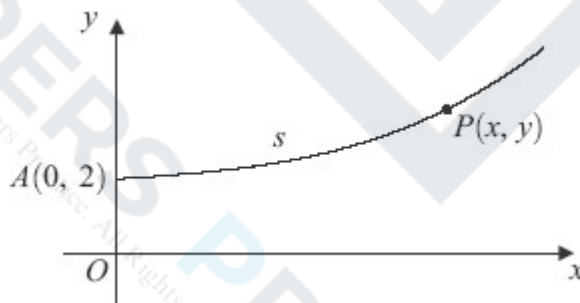
(Total 12 marks)

Q6.

The diagram shows a curve which starts from the point A with coordinates $(0, 2)$. The curve is such that, at every point P on the curve,

$$\frac{dx}{dy} = \frac{1}{2}s$$

where s is the length of the arc AP .



(a) (i) Show that

$$\frac{ds}{dx} = \frac{1}{2}\sqrt{4+s^2}$$

(3)

(ii) Hence show that

$$s = 2 \sinh \frac{x}{2}$$

(4)

(iii) Hence find the cartesian equation of the curve.

(3)

- (b) Show that

$$y^2 = 4 + s^2$$

(2)

(Total 12 marks)

Q7.

- (a) Given that $y = \ln \tanh \frac{x}{2}$, where $x > 0$, show that

$$\frac{dy}{dx} = \operatorname{cosech} x$$

(6)

- (b) A curve has equation $y = \ln \tanh \frac{x}{2}$, where $x > 0$. The length of the arc of the curve between the points where $x = 1$ and $x = 2$ is denoted by s .

- (i) Show that

$$s = \int_1^2 \coth x \, dx$$

(2)

- (ii) Hence show that $s = \ln(2 \cosh 1)$.

(4)

(Total 12 marks)

Q8.

- (a) Given that $y = \operatorname{sech} t$, show that:

(i) $\frac{dy}{dt} = -\operatorname{sech} t \tanh t$

(3)

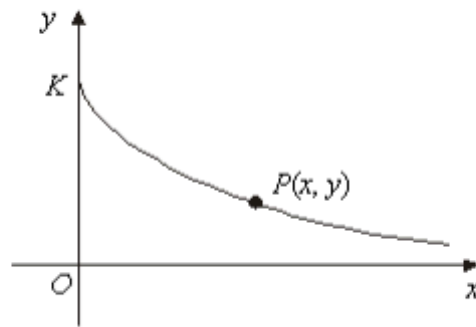
(ii) $\left(\frac{dy}{dt}\right)^2 = \operatorname{sech}^2 t - \operatorname{sech}^4 t$.

(2)

- (b) The diagram shows a sketch of part of the curve given parametrically by

$$x = t - \tanh t$$

$$y = \operatorname{sech} t$$



The curve meets the y -axis at the point K , and $P(x, y)$ is a general point on the curve. The arc length KP is denoted by s . Show that:

- (i) $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \tanh^2 t;$ (4)
- (ii) $s = \ln \cosh t;$ (3)
- (iii) $y = e^{-s}.$ (2)
- (c) The arc KP is rotated through 2π radians about the x -axis. Show that the surface area generated is $2\pi (1 - e^{-s})$ (4)
- (Total 18 marks)

Q9.

A curve has equation $y = 4\sqrt{x}$.

- (a) Show that the length of arc s of the curve between the points where $x = 0$ and $x = 1$ is given by

$$s = \int_0^1 \sqrt{\frac{x+4}{x}} \, dx$$

(4)

- (b) (i) Use the substitution $x = 4 \sinh^2 \theta$ to show that

$$\int \sqrt{\frac{x+4}{x}} \, dx = \int 8 \cosh^2 \theta \, d\theta$$

(5)

- (ii) Hence show that $s = 4 \sinh^{-1} 0.5 + \sqrt{5}$

(6)
(Total 15 marks)

Q10.

(a) Use the definitions

$$\sinh \theta = \frac{1}{2}(e^{\theta} - e^{-\theta}) \quad \text{and} \quad \cosh \theta = \frac{1}{2}(e^{\theta} + e^{-\theta})$$

to show that:

(i) $2 \sinh \theta \cosh \theta = \sinh 2\theta;$

(2)

(ii) $\cosh^2 \theta + \sinh^2 \theta = \cosh 2\theta.$

(3)

(b) A curve is given parametrically by

$$x = \cosh^3 \theta, \quad y = \sinh^3 \theta$$

(i) Show that

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \frac{9}{4} \sinh^2 2\theta \cosh 2\theta$$

(6)

(ii) Show that the length of the arc of the curve from the point where $\theta = 0$ to the point where $\theta = 1$ is

$$\frac{1}{2} \left[(\cosh 2)^{\frac{3}{2}} - 1 \right]$$

(6)

(Total 17 marks)